

Business Plan

App name: Gem Finder

One sentence description :

What is a gem? The gem is a good view place that has high scores and low busyness at some point.

Gem Finder is a software that helps you discover gems hidden beneath the “earth”.

Let's use the gem finder together and start our exploration!

What lead me to do this:

Existing travel apps such as Tripadvisor mainly rely on attraction ratings and reviews for recommendations. However, the actual experience of visiting each attraction is closely related to the time of day and the busyness of the place.

Currently available products don't recommend both highly-rated and fun attractions, or suggest times when they are less busy. This prompted me to think about incorporating the travel rating system with information on the busyness levels of the day and the following day, to provide recommendations for highly-rated attractions that are less busy level on either the current day or the next of the place.

The core problem to solve:

How can users find time periods with high ratings but low busyness level to have a better user experience at tourist attractions?

(The users can get recommended travel destinations and busyness level predictions from the system based on the preferred place category.)

Target Users:

Tourists have preferences for certain types of attractions and hope to have a better travel experience.

Competitors:

Tripadvisor; Booking; Dianping

	Gem Finder	TripAdvisor	Booking	Dianping
Core discovery				
Attraction listing by category	✓	✓	✓	✓
Attraction rating display	✓	✓	◦ Hotels only	✓
User-generated reviews	◦ Rating only	✓	✓	✓
Search & keyword filter	◦ Category filter	✓	✓	✓
Busyness level & timing intelligence				
Real-time crowd level	✓ Core feature	✗	✗	✗

Crowd forecast: today & tomorrow	✓ 30min slots	X	X	X
Crowd heat map by zone	✓ With time slider	X	X	X
Opening status (live)	✓	◦ Static hours only	◦ Static hours only	◦ Static hours only
"Gem period" highlight (low crowd + open)	✓ Auto-highlighted	X	X	X
Gem recommendation	✓ High rating + low crowd	X	X	X
Accessibility				
Mobility accessibility filter	✓ Dedicated filter	◦ Mentioned in reviews only	◦ Hotels only	X
High-contrast UI mode	✓	X	X	X
User system & engagement				
User registration & login	✓	✓	✓	✓
Favorite / save attractions	✓	✓	✓	✓
Check-in at attractions	✓ With gem box	X	X	✓
Gamified check-in system	✓ 2.5D gem collection	X	X	◦ Points only
Personal check-in map	✓	X	X	◦ Limited
User profile page	✓	✓	✓	✓
Smart features				
Weather chatbot / outfit advice	✓	X	X	X
Admin & platform				
Admin user management	✓ Role + status control	✓	✓	✓
Geographic focus	Manhattan only	Global	Global	China-focused
Target user	Tourists in Manhattan	Global travelers	Hotel bookers	Urban diners & locals

Tripadvisor is a review-driven global discovery platform, Booking focuses on hotel bookings, Dianping is a Chinese review platform, and Gem Finder is a Manhattan-specific tool that integrates "busyness flow × ratings."

In the entire category of busyness flow and timing (real-time busyness level, prediction, heatmaps, and gem time recommendations), all three competitors are not provided, which is Gem Finder's biggest differentiating advantage.

In terms of user interaction, the game style gem box check-in system is a unique function of Gem Finder. While Dianping has a points system, it lacks the visual appeal of Gem Finder's design.

(Google Maps tells you "where this place is," while Gem Finder tells you "is it worth going there now?")

Google Maps' visitor count display is based on passive historical data ("Saturdays are usually busy"), while Gem Finder's core value lies in recommendations—when a highly-rated attraction is currently less crowded and open, the system will tell you "now is the best time to go," a Gem Period logic that competitors lack.)

Architecture and Technology Stack:

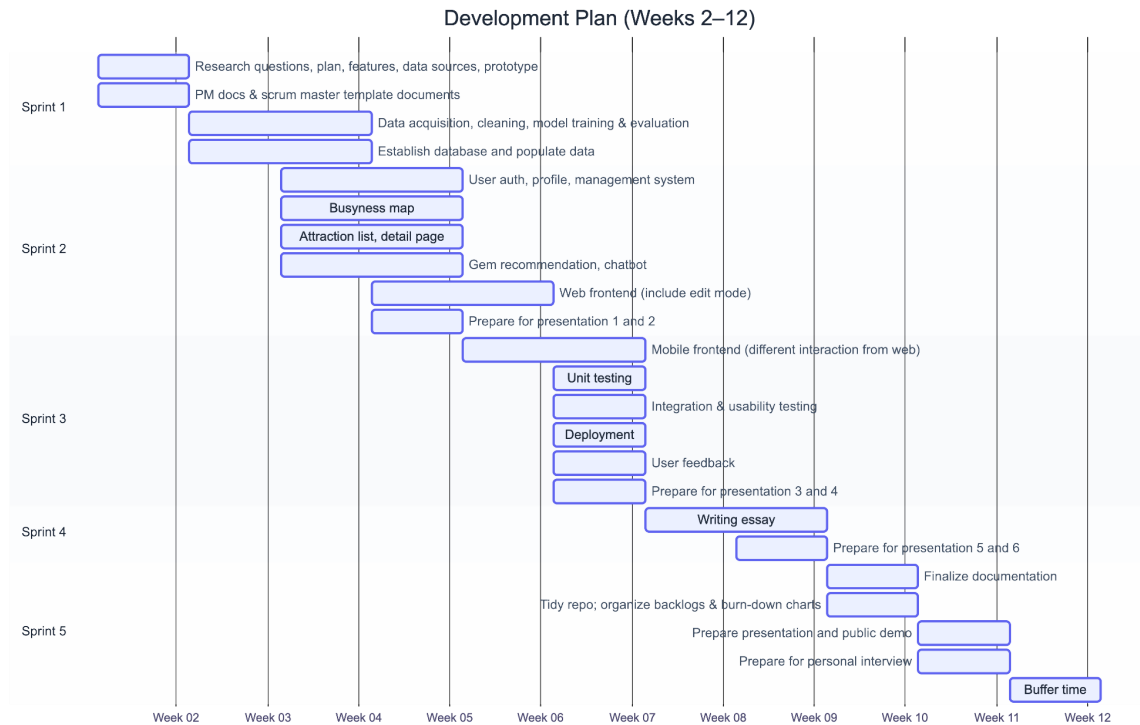
Layer	Technology	Category	Notes
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Frontend web	React	Frontend	(Need to further learn)
Frontend mobile	React Native	Frontend	(Need to learn)
Backend	Spring Boot	Backend	JPA / MyBatis
Model serving	Flask	ML	Python ML models
Database	MySQL	Database	Primary storage
Cache	Redis	Database	(Need to further learn)
Local service	Ollama	ML	(Might not use — Server might not be sufficient — Need further investigation)
Deployment	Docker Compose	Deployment	
Documentation	Notion / Google Docs	Infra	

Development Plan: Timeline and Milestones, Risks and Mitigations:

Sprint	Week	Focus	Tasks	Risk	Mitigation
Sprint 1	Week 2	Planning	• Research questions, plan, features, data sources, prototype • PM documents & scrum master template documents	Low	Work solidly
	Week 3	ML + DB	• Data acquisition, cleaning, model training & evaluation • Establish database and populate data	High The initial data acquisition and model training processes involve a high degree of uncertainty.	Therefore, I reduced the time spent on product-related aspects in the early stages, moved the timeline forward, and allowed for contingency.
Sprint 2	Week 4	Backend	• User auth, profile, management system • Busyness map • attraction list, detail page • Gem recommendation, chatbot	Medium Heavy feature load in one week	Work solidly

	Week 5	Web frontend	• Web frontend (include edi mode) • Prepare for presentation 1 and 2	Low React learning Presentation prep competes with dev time	presentation preparation for 1 day
Sprint 3	Week 6	Mobile frontend	• Mobile frontend (different interaction from web)	Medium React Native learning	Work solidly
	Week 7	Testing	• Unit testing • Integration & usability testing • Deployment • User feedback • Prepare for presentation 3 and 4	Medium presentation prep overlap	Work solidly
Sprint 4	Week 8	Essay	• Writing essay	High Previously, I only wrote Harvard-style papers; I need to learn academic writing and literature review, which requires a significant amount of time.	Therefore, I need to move quickly in the early stages to allow time for writing (an 8-page IEEE paper might require 4000 words; I can usually write 500 words a day, so this needs to be addressed with a dedicated amount of time).
	Week 9	Essay	• Writing essay • Prepare for presentation 5 and 6	High Essay + presentation prep overlap	Work solidly
Sprint 5	Week 10	Docs	• Finalize documentation • Tidying up repo documents; Organize sprint backlogs & product backlogs & burn-down charts	Low	Work solidly
	Week 11	Demo	• Prepare presentation and public demo • Prepare for personal interview	Low	Work solidly
	Week 12	Buffer	• Buffer time	Low	Buffer — absorb delays from essay



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Writing IEEE format papers requires learning how to write and cite. Previously, I only wrote Harvard-style papers; I need to learn academic writing and literature review, which requires a significant amount of time. Therefore, I need to move quickly in the early stages to allow time for writing (an 8-page IEEE paper might require 4000 words; I can usually write 500 words a day, so this needs to be addressed with a dedicated amount of time).

In summary, the initial data acquisition and ML model training are the most demanding and challenging parts, and also the ones that worry me the most. Secondly, writing the paper requires a significant amount of time. Therefore, for me, shifting the overall timeline forward, pushing each part forward as much as possible, and allowing for a relatively large buffer at the end might give me more confidence and ensure the project's successful completion.

Plan Budget:

	Feature/Task	Time Est.	+Buffer(30%)	Total Time	Hourly Rate	Human Cost (Time x Rate)	Tools/ Services	Service Cost €	Total Cost €
1	Requirements analysis, competitor analysis, mockups	20	6	26	€ 20/hr	€520.00	Figma/Stitch	€0.00	€520.00
2	Data Fetching and Cleaning	25	7.5	32.5	€ 20/hr	€650.00	NYC OpenData	€0.00	€650.00

3	ML Model (busyness predicting)	22	6.6	28.6	€ 20/hr	€572.00	scikit-learn / pandas/ Flask	€0.00	€572.00
4	User auth system	15	4.5	19.5	€ 20/hr	€390.00	-	€0.00	€390.00
5	User profile pages	12	3.6	15.6	€ 20/hr	€312.00	-	€0.00	€312.00
6	Management system	12	3.6	15.6	€ 20/hr	€312.00	-	€0.00	€312.00
7	Data visualisation — busyness map	20	6	26	€ 20/hr	€520.00	OpenStreetMap	€0.00	€520.00
8	Attraction list & detail pages	16	4.8	20.8	€ 20/hr	€416.00		€0.00	€416.00
9	Gem recommendation feature	14	4.2	18.2	€ 20/hr	€364.00		€0.00	€364.00
10	GenAI Chatbot	15	4.5	19.5	€ 20/hr	€390.00	Ollama	€0.00	€390.00
11	EDI — high contrast mode	6	1.8	7.8	€ 20/hr	€156.00	-	€0.00	€156.00
12	Web frontend	30	9	39	€ 20/hr	€780.00	React	€0.00	€780.00
13	Mobile responsiveness & UI adaptation & deployment	25	7.5	32.5	€ 20/hr	€650.00	React Native	€0.00	€650.00
14	Testing, Debugging	12	3.6	15.6	€ 20/hr	€312.00	Testing tools	€0.00	€312.00
15	Deployment	15	4.5	19.5	€ 20/hr	€390.00	Docker Compose, Hetzner	€10.00	€400.00
16	Documentation & Presentation	10	3	13	€ 20/hr	€260.00	Notion/ Google docs	€0.00	€260.00
	Total	269	80.7	349.7	-	€6,994.00		€10.00	€7,004.00